

Операционные системы

Поиск файлов. Перенаправление ввода-вывода. Просмотр запущенных процессов

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Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

- 1 Изучить перенаправление ввода-вывода
- 2 Изучить работу фильтров
- 3 Изучить команду поиска
- 4 Ознакомиться с управлением процессами
- 5 Ознакомиться с командами `df` `du`

2. Процесс выполнения лабораторной работы

```
miradov@miradov:~$ ls /etc/ > file.txt
miradov@miradov:~$ ls >> file.txt
miradov@miradov:~$ cat file.txt
abrt
adjtime
aliases
alsa
alternatives
anaconda
anacrontab
anthy-unicode.conf
asound.conf
at.deny
audit
authselect
avahi
bash_completion.d
bashrc
bindresvport.blacklist
binfmt.d
bluetooth
brlapi.key
brltty
brltty.conf
ceph
chromium
```

Рисунок 1: Запись в файл

```
miradov@miradov:~$ grep .conf file.txt > conf.txt
miradov@miradov:~$ cat conf.txt
anthy-unicode.conf
asound.conf
brltty.conf
chrony.conf
cpupower-service.conf
dconf
dnsmasq.conf
dracut.conf
dracut.conf.d
fprintd.conf
fuse.conf
host.conf
idmapd.conf
kdump.conf
krb5.conf
krb5.conf.d
ld.so.conf
ld.so.conf.d
libaudit.conf
libuser.conf
locale.conf
logrotate.conf
makedumpfile.conf.sample
```

Рисунок 2: Поиск расширения .conf


```
object-personal/stage01/report/bib/cite.bib
/home/miradov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/pr
object-personal/stage02/report/_resources/csl
/home/miradov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/pr
object-personal/stage02/report/bib/cite.bib
/home/miradov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/pr
object-personal/stage03/report/_resources/csl
/home/miradov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/pr
object-personal/stage03/report/bib/cite.bib
/home/miradov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/pr
object-personal/stage04/report/_resources/csl
/home/miradov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/pr
object-personal/stage04/report/bib/cite.bib
/home/miradov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/pr
object-personal/stage05/report/_resources/csl
/home/miradov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/pr
object-personal/stage05/report/bib/cite.bib
/home/miradov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/pr
object-personal/stage06/report/_resources/csl
/home/miradov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/pr
object-personal/stage06/report/bib/cite.bib
/home/miradov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/pr
resentation/report/_resources/csl
/home/miradov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/pr
resentation/report/bib/cite.bib
/home/miradov/git-extended/.git/hooks/commit-msg.sample
/home/miradov/git-extended/.git/objects/09/c20e275115c93ec2c35405a02799e57bd1f999
/home/miradov/git-extended/.git/objects/85/c8151cfa6c92982abeae22d23b154803d0654e
/home/miradov/git-extended/.git/objects/c5
/home/miradov/git-extended/.git/config
/home/miradov/conf.txt
miradov@miradov:~$
```

Рисунок 3: Поиск файлов

```
/etc/foomatic/hashes.d/hashes.new
find: '/etc/libvirt': Permission denied
/etc/hp
/etc/hp/hplip.conf
/etc/httpd
/etc/httpd/conf/httpd.conf
/etc/libibverbs.d/hfi1verbs.driver
/etc/libibverbs.d/hns.driver
find: '/etc/lvm/archive': Permission denied
/etc/logrotate.d/httpd
find: '/etc/lvm/backup': Permission denied
find: '/etc/lvm/cache': Permission denied
find: '/etc/lvm/devices': Permission denied
find: '/etc/nftables': Permission denied
find: '/etc/openvpn/client': Permission denied
find: '/etc/openvpn/server': Permission denied
/etc/nvme/hostnqn
/etc/nvme/hostid
find: '/etc/pki/rsyslog': Permission denied
find: '/etc/polkit-1/localauthority': Permission denied
find: '/etc/polkit-1/rules.d': Permission denied
find: '/etc/sos/cleaner': Permission denied
/etc/sane.d/dll.d/hpaio
/etc/sane.d/hp.conf
/etc/sane.d/hp3900.conf
/etc/sane.d/hp4200.conf
/etc/sane.d/hp5400.conf
/etc/sane.d/hpsj5s.conf
```

Рисунок 4: Поиск файлов

```
miradov@miradov:~$  
miradov@miradov:~$ find /etc -name "h*" -print | less  
miradov@miradov:~$  
miradov@miradov:~$  
miradov@miradov:~$ find ~ -name "log*" > logfile &  
[1] 20238  
miradov@miradov:~$  
[1]+  Done                  find ~ -name "log*" > logfile  
miradov@miradov:~$ rm logfile  
miradov@miradov:~$
```

Рисунок 5: Фоновый запуск процесса

```
miradov@miradov:~$  
miradov@miradov:~$ gedit &  
[1] 20280  
miradov@miradov:~$ ps | grep gedit  
20280 pts/1    00:00:00 gedit  
miradov@miradov:~$ kill 20280  
[1]+  Terminated                  gedit  
miradov@miradov:~$  
miradov@miradov:~$
```

Рисунок 6: Фоновый запуск и завершение процесса

```
DF(1)                                User Commands                                DF(1)

NAME
    df - report file system space usage

SYNOPSIS
    df [OPTION]... [FILE]...

DESCRIPTION
    This manual page documents the GNU version of df.  df displays the amount
    of space available on the file system containing each file name argument.
    If no file name is given, the space available on all currently mounted
    file systems is shown.  Space is shown in 1K blocks by default, unless the
    environment variable POSIXLY_CORRECT is set, in which case 512-byte blocks
    are used.

    If an argument is the absolute file name of a device node containing a
    mounted file system, df shows the space available on that file system
    rather than on the file system containing the device node.  This version
    of df cannot show the space available on unmounted file systems, because
    on most kinds of systems doing so requires non-portable intimate knowledge
    of file system structures.

OPTIONS
    Show information about the file system on which each FILE resides, or all
    file systems by default.

    Mandatory arguments to long options are mandatory for short options too.

    -a, --all
        include pseudo, duplicate, inaccessible file systems

    -B, --block-size=SIZE
        scale sizes by SIZE before printing them; e.g., '-BM' prints sizes
        in units of 1,048,576 bytes; see SIZE format below

    --direct
        Manual page df(1) line 1 (press h for help or q to quit)
```

Рисунок 7: Справка по команде df

```
DU(1)                                User Commands                                DU(1)

NAME
    du - estimate file space usage

SYNOPSIS
    du [OPTION]... [FILE]...
    du [OPTION]... --files0-from=F

DESCRIPTION
    Summarize device usage of the set of FILES, recursively for directories.

    Mandatory arguments to long options are mandatory for short options too.

    -0, --null
        end each output line with NUL, not newline

    -a, --all
        write counts for all files, not just directories

    --apparent-size
        print apparent sizes rather than device usage; although the apparent size is usually smaller, it may be larger due to holes in ('sparse') files, internal fragmentation, indirect blocks, and the like

    -B, --block-size=SIZE
        scale sizes by SIZE before printing them; e.g., '-BM' prints sizes in units of 1,048,576 bytes; see SIZE format below

    -b, --bytes
        equivalent to '--apparent-size --block-size=1'

    -c, --total
        produce a grand total

    -D, --dereference-args
        follow symbolic links that are arguments to du

Manual page du(1) line 1 (press h for help or q to quit)
```

Рисунок 8: Запуск команды df

```
miradov@miradov:~$ df
Filesystem      1K-blocks    Used Available Use% Mounted on
/dev/nvme0n1p3 155186176 59773056 90107824 40% /
devtmpfs        4011280      0    4011280  0% /dev
tmpfs           4035656      92    4035564  1% /dev/shm
tmpfs           1614264     1956   1612308  1% /run
tmpfs           1024         0       1024    0% /run/credentials/systemd-journald
.service
tmpfs           4035656    132940   3902716  4% /tmp
/dev/nvme0n1p3 155186176 59773056 90107824 40% /home
/dev/nvme0n1p2 1992552    518800   1352512 28% /boot
/dev/loop2      49280     49280      0 100% /var/lib/snapd/snap/snapd/25935
/dev/loop0      68480     68480      0 100% /var/lib/snapd/snap/core24/1349
/dev/loop1     115712    115712      0 100% /var/lib/snapd/snap/hugo/25676
tmpfs           1024         0       1024    0% /run/credentials/systemd-resolved
.service
tmpfs           807128      200    806928  1% /run/user/1086
miradov@miradov:~$
```

Рисунок 9: Справка по команде du

```
4      ./git-extended/.git/objects/4f
4      ./git-extended/.git/objects/0c
4      ./git-extended/.git/objects/f3
4      ./git-extended/.git/objects/5d
4      ./git-extended/.git/objects/aa
4      ./git-extended/.git/objects/0d
8      ./git-extended/.git/objects/c5
76     ./git-extended/.git/objects
8      ./git-extended/.git/logs/refs/heads
8      ./git-extended/.git/logs/refs/remotes/origin
8      ./git-extended/.git/logs/refs/remotes
16     ./git-extended/.git/logs/refs
20     ./git-extended/.git/logs
212    ./git-extended/.git
220    ./git-extended
0      ./monthly
0      ./reports/monthly/monthly
0      ./reports/monthly
0      ./reports
4      ./ski.plases/equipment
0      ./ski.plases/plans
4      ./ski.plases
0      ./australia
0      ./play/games/play
0      ./play/games
0      ./play
824708 .
miradov@miradov:~$
```

Рисунок 10: Запуск команды du


```
/home/miradov/git-extended/.git/objects/e1
/home/miradov/git-extended/.git/objects/85
/home/miradov/git-extended/.git/objects/bb
/home/miradov/git-extended/.git/objects/50
/home/miradov/git-extended/.git/objects/3f
/home/miradov/git-extended/.git/objects/4f
/home/miradov/git-extended/.git/objects/0c
/home/miradov/git-extended/.git/objects/f3
/home/miradov/git-extended/.git/objects/5d
/home/miradov/git-extended/.git/objects/aa
/home/miradov/git-extended/.git/objects/0d
/home/miradov/git-extended/.git/objects/c5
/home/miradov/git-extended/.git/logs
/home/miradov/git-extended/.git/logs/refs
/home/miradov/git-extended/.git/logs/refs/heads
/home/miradov/git-extended/.git/logs/refs/remotes
/home/miradov/git-extended/.git/logs/refs/remotes/origin
/home/miradov/monthly
/home/miradov/reports
/home/miradov/reports/monthly
/home/miradov/reports/monthly/monthly
/home/miradov/ski.places
/home/miradov/ski.places/equipment
/home/miradov/ski.places/plans
/home/miradov/australia
/home/miradov/play
/home/miradov/play/games
/home/miradov/play/games/play
miradov@miradov:~$
```

Рисунок 11: Поиск директорий

3. Выводы по проделанной работе

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.